



Sri Lanka Institute of Information Technology

M.Sc. in Artificial Intelligence

Final Examination
Year 1, Semester 1 (2025)

IT5022 – Fundamentals of Machine Learning

Duration: 2 Hours

November 2025

Instructions to Candidates:

- ◆ This paper has 4 questions.
- ◆ Answer all questions in the booklet given.
- ◆ The total marks for the paper are 100.
- ◆ This paper contains 10 pages, including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including mobile phones are not allowed.
- ◆ Non-programmable calculators allowed.
- ◆ **One** A4-size reference sheet, written or printed on both sides is allowed.

Question 1**(25 marks)**

a) An online retail company is building a machine learning system to predict customer churn, recommend products, and segment customer groups based on behavior. They have large datasets, some with labels (churned vs. not churned), and others with only transaction histories.

i. Which machine learning approach should be used to predict customer churn?

State the type of learning and give one algorithm.

(2 marks)

ii. The marketing team wants to group customers based on purchasing patterns without predefined labels. Which learning technique should they apply? State the type of learning and name one algorithm.

(2 marks)

iii. If only 10% of the customer data has churn labels and 90% is unlabeled, which learning approach should the team use to train the model using both? State the approach and give a brief reason using one sentence.

(2 marks)

b) A farmer plans to cultivate two types of crops: wheat (W) and corn (C).

- Each acre of wheat earns a profit of \$200.
- Each acre of corn earns a profit of \$300.
- The farmer has the following resource limitations:
 - Land: maximum 100 acres.
 - Water: wheat requires 3 units of water per acre; corn requires 5 units; total available water is 400 units.
 - Labor: maximum 240 labor hours; wheat requires 2 hours per acre, corn requires 4 hours.

The goal is to maximize profit.

Objective Function:

Maximize: $Z = 200W + 300C$

Subject to constraints:

$W + C \leq 100$ (land constraint)

$3W + 5C \leq 400$ (water constraint)

$2W + 4C \leq 240$ (labor constraint)

$W \geq 0, C \geq 0$

i. Is this a problem that can be used using a Machine Learning algorithm or an Optimization algorithm? Justify your answer with one sentence.

(2 marks)

ii. Out of the algorithms linear programming, linear regression and random forest, which algorithm may be most optimal for solving the above problem. Explain with one sentence. (2 marks)

iii. The objective function can be improved further with predicted rainfall information, which is stored as a continuous value that varies over time. Identify a category of algorithms that can be used to predict the rainfall. Name one algorithm under this category. (2 marks)

c) A real estate company is training a machine learning model to predict house prices.

- Dataset includes:

- Floor area (sq.ft): 500 to 4000
- Number of bedrooms: 1 to 5
- Age of house (years): 0 to 50
- The model uses the following hypothesis function:

$$h(\theta) = \theta_0 + \theta_1 \times \text{Area} + \theta_2 \times \text{Bedrooms} + \theta_3 \times \text{Age}$$

The cost function to be minimized is:

$$J(\theta) = (1/2m) \sum (h(\theta^i) - y^i)^2$$

To prevent overfitting, the regularized cost function becomes:

$$J(\theta) = (1/2m) \sum (h(\theta^i) - y^i)^2 + (\lambda/2m) \sum \theta_j^2$$

i. Which regression model is used according to the hypothesis function? Justify the answer with one sentence. (2 marks)

ii. Based on the scale of the features given, what preprocessing technique should be applied to improve gradient descent convergence? (2 marks)

iii. In the regularized cost function, what is the role of λ (explain with one sentence)? (2 marks)

iv. Which optimization algorithm is typically used to minimize both versions of the cost function? Briefly explain how it works with one sentence. (2 marks)

d) A data scientist builds two regression models (Model A and Model B) to predict monthly electricity consumption (in kWh) based on household data.

Household	Actual	Pred A	Pred B
1	420	410	430
2	380	390	370
3	450	440	410
4	400	415	395
5	460	450	440

i. Compute the Mean Squared Error (MSE) for both Model A and Model B. Which model performs better? (2 marks)

ii. Find the R^2 score for both models and briefly interpret the result for the better model.

(2 marks)

iii. If Model A has a slightly higher MSE than Model B but a higher R^2 , briefly explain (using one sentence) how that can happen. (1 mark)

Question 2

(25 marks)

a) A hospital is developing a machine learning model to predict whether a patient has diabetes (yes or no) based on features like age, BMI, blood pressure, and glucose level. The model uses logistic regression.

The prediction is made using the logistic function:

$$h(\theta) = 1 / (1 + e^{-\theta^T x})$$

The regularized cost function used during training is:

$$J(\theta) = -(1/m) \sum [y \log(h(\theta)) + (1 - y) \log(1 - h(\theta))] + (\lambda/2m) \sum \theta_j^2$$

i. The hospital observes that the model is consistently predicting poorly on both training and test data (underfitting). Based on the regularized cost function above, what change should be made to improve the model's learning? (2 marks)

ii. The hospital sets a classification threshold of 0.5. For a new patient, the model outputs $h(\theta) = 0.85$. What is the predicted class for this patient? Justify using one sentence.

(2 marks)

iii. For one patient, the true label is diabetes positive (label 1). The model predicts a probability of 0.9. Calculate the cost for this data point (round to 3 decimal places), using the given cost function. Write the intermediate steps in the calculation. (2 marks)

b) A marketing company is building a predictive model using both Decision Tree and Random Forest algorithms to predict whether customers will buy a new product. The dataset consists of 15 customers:

- 10 customers said 'yes', 5 customers said 'no'.

The entropy of the full dataset is:

$$\text{Entropy}(D) = - (10/15)\log_2(10/15) - (5/15)\log_2(5/15) = 0.918 \text{ bits}$$

After splitting on Age, the data is divided as:

- Youth: 6 samples (2 yes, 4 no)

- Middle-aged: 5 samples (5 yes, 0 no)

- Senior: 4 samples (3 yes, 1 no)

The company is concerned about overfitting since the dataset is small.

i. Calculate the entropy for the Middle-aged group. Write the intermediate calculations.

(2 marks)

ii. Compute the weighted average entropy after splitting by Age. Write the intermediate steps in the calculation.

Individual sub-group entropies:

- Youth: Entropy ≈ 0.918 bits

- Senior: Entropy ≈ 0.811 bits

(2 marks)

iii. The company finds that the decision tree perfectly fits the training data but performs poorly on new customers. How can pruning help reduce this problem (explain with one sentence)? (2 marks)

iv. The company starts with 10 trees in the Random Forest model and increases it to 100 trees. Assuming the feature set and data remain the same, explain how this change affects prediction accuracy for future customers. (2 marks)

c) A financial institution is building a credit risk prediction system. Based on customer features like income, debt ratio, and payment history, the goal is to classify customers into two groups: "Low Risk" and "High Risk".

The dataset shows some overlap: a few low-risk customers behave like high-risk ones, and vice versa, such that it may not be perfectly and linearly separable. The institution decides to use a Support Vector Machine (SVM) to build the classifier.

i. Should the company use a hard-margin or soft-margin SVM for this problem? Briefly explain your choice based on the nature of the data. (2 marks)

ii. The institution tests two SVM models:

- Model A: $C = 100$

- Model B: $C = 0.1$

Explain which model is more likely to overfit. Briefly explain with one sentence.

(2 marks)

iii. If the company observes that even the technique used in c.i struggles to separate the customers in the current feature space, what technique can they apply to better separate the classes by transforming the dataset to a different feature space? Justify your answer with one sentence. (2 marks)

d) A machine learning engineer builds a binary classifier to detect spam emails. The model outputs a probability score between 0 and 1 for each email being spam. On a test set of 10,000 emails (4,000 spam and 6,000 non-spam), the engineer evaluates the model using ROC-AUC.

i. Briefly explain (using 1-2 sentences) what the ROC curve represents and how the AUC value is interpreted in this context. (2 marks)

ii. Model X has an $AUC = 0.93$, while Model Y has $AUC = 0.84$. Both show similar accuracy ($\approx 89\%$). Briefly explain (using one sentence) why Model X is still better for deployment. (1 mark)

iii. After deployment, the engineer notices that the AUC on real user data drops from 0.93 to 0.78. Give two possible reasons for this and suggest one action to address it. (2 marks)

Question 3**(25 marks)**

a) A hospital is developing a machine learning model to detect a rare form of cancer. Out of 5,000 patients, only 100 have the cancer (positive class), and 4,900 do not (negative class). The model produced the following confusion matrix on test data:

	Predicted Cancer	Predicted No Cancer
Actual Cancer	80	20
Actual No Cancer	200	4700

- i. Calculate precision. Use the formula: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$. Then briefly explain its meaning using one sentence. (2 marks)
- ii. Calculate accuracy. Use the formula: $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$. Then briefly explain its meaning using one sentence. (2 marks)
- iii. Based on the above results, is accuracy alone sufficient for evaluating this model? Briefly explain the reason for your answer using one sentence. (2 marks)
- iv. Name one additional metric that may be useful to evaluate the above model. Briefly explain how it operates using one sentence. (2 marks)
- v. How does class imbalance affect this problem? Briefly explain one method to address it. (2 marks)

b) A supermarket chain has collected customer data to understand purchasing patterns. The dataset includes:

- Age (numerical)
- Monthly Spending (numerical)
- Product Preference (categorical: "Organic", "Imported", "Discounted")
- Payment Type (categorical: "Credit Card", "Cash", "Mobile Payment")

The marketing team wants to segment customers for personalized promotions using unsupervised learning.

- i. Before applying clustering, the team needs to decide how many clusters (K) to use. Name one method they can use to select a good value for K and briefly explain how it operates using one sentence. (2 marks)
- ii. The team applies K-Prototypes for clustering. Briefly explain how K-Prototypes may be more suitable for clustering the above data than K-means. (2 marks)
- iii. The team wants to visualize the customer segments in 2D. Explain using one sentence how PCA can be useful for this task. (1 mark)

c) A shopping mall collects data on customer movement throughout the mall to understand walking patterns. The data includes x-y coordinates recorded every few seconds as customers move around. The mall's data science team wants to cluster these movement patterns to identify common shopping behaviors.

Because the data has many irregular walking paths, some areas are dense (food courts, entrances), while others are sparse (service corridors). The team applies DBSCAN but also considers hierarchical clustering for certain parts of the analysis.

- i. How can DBSCAN be used to detect unusual customer paths (outliers) in this scenario? Explain using one sentence. (2 marks)
- ii. Explain one advantage DBSCAN offers compared to K-Means when analyzing these irregular movement patterns. (2 marks)
- iii. The team now wants to analyze customer groups based on shopping preferences (age, income, preferred stores). In this new case, briefly explain (using 1 sentence) why hierarchical clustering may be useful. (1 mark)

d) A data scientist is training a Random Forest Regressor to predict house prices in Colombo using features such as area, number of bedrooms, distance to city center, and age of the house. To improve model performance, she decides to perform hyperparameter tuning.

Key hyperparameters under consideration:

`n\_estimators` (number of trees)

`max\_depth` (maximum depth of each tree)

`min\_samples\_split` (minimum samples required to split a node)

- i. Briefly explain (using one sentence) why hyperparameter tuning is important in this model and what could happen if default values are used. (2 marks)

ii. The scientist tests combinations of parameters using Grid Search and Random Search. Briefly explain the main difference between the two methods, and when Random Search might be preferred. (2 marks)

iii. After tuning, the model with `'n_estimators=200'`, `'max_depth=15'`, and `'min_samples_split=4'` shows a small validation improvement but takes $3\times$ longer to do the inference or prediction. Briefly explain (using 1-2 sentences) what should the data scientist consider before deploying it? (1 mark)

Question 4

(25 marks)

a) A cloud engineer monitors the response time (in milliseconds) of servers in a data center. Under normal operating conditions, response times follow a Gaussian distribution with:

$$\mu = 250, \sigma = 30$$

A new batch of readings from one server shows unusually high response times, raising concern about anomalies.

i. Briefly explain how the Gaussian model is used for anomaly detection in this context using 1-2 sentences. (2 marks)

ii. Compute the probability density for a reading of 320 ms and comment on whether it's likely an anomaly. (2 marks)

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

iii. Briefly describe how you would set the anomaly threshold (ϵ) and tune it for better performance. (2 marks)

iv. If the engineer later adds network traffic load as a second feature that is correlated to the response time, explain how the Gaussian model changes. (2 marks)

b) A streaming platform wants to improve its movie recommendation system using collaborative filtering. They have a dataset of user-movie ratings ranging from 1 to 5. The team decides to use user-based collaborative filtering (UBCF) to suggest unseen movies to users.

- i. Briefly explain (using 1-2 sentences) the main idea behind collaborative filtering and how it differs from a content-based approach. (2 marks)
- ii. If User A and User B both rated Inception, Interstellar, and The Matrix similarly, how would the system use this information to recommend new movies to User A? Briefly explain using 1-2 sentences. (1 mark)
- iii. The team notices that users who rate very few movies are causing noisy predictions. What approach can reduce this problem? (2 marks)
- iv. Briefly explain one limitation of collaborative filtering and one way to mitigate it in real-world systems. (2 marks)

c) A logistics company deploys an autonomous robot to move packages inside a warehouse. The warehouse is divided into a 5×5 grid where each cell represents a location. Some cells contain obstacles, while one cell represents the delivery zone (goal). The robot learns through Q-learning how to navigate efficiently from its starting position to the delivery zone.

- i. Briefly explain how the robot uses the Q learning algorithm to learn the optimal path. (2 marks)
- ii. Suppose the learning rate (α) is set too high, and the discount factor (γ) is very low. Briefly describe the effect of these settings on the robot's learning behavior. (2 marks)
- iii. After training, the robot performs well in the training grid but fails when a few obstacles are moved. Briefly explain why this happens and suggest one improvement. (2 marks)

d) A bank develops a machine learning model to predict whether applicants should be approved or denied for personal loans. After deployment, auditors find that the approval rates differ between male and female applicants, raising concerns about bias and fairness.

- i. Briefly explain the difference between Demographic Parity and Equalized Odds in evaluating this model's fairness. (2 marks)

ii. Suppose the bank's model has:

Male approval rate = 72%, Female approval rate = 58%,

True positive rate for males = 80%, for females = 79%.

Which fairness metric is violated, and what practical action should the bank take?

(2 marks)